ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C2 · PUSHES AND PULLS

Push or pull?

★☆☆ Grade 2

Learning goals

- Sort pushes and pulls.
- Name forces.

1 Push or pull? [6]

SORT THESE

pushing a pram **kicking a ball** **opening a drawer** **pulling a rope**
pressing a button **flying a kite**

Push	Pull
kicking a ball	opening a drawer
pushing a pram	flying a kite
pressing a button	pulling a rope

2 What does the force do? Draw a line. [4]

Draw a line to match each pair, or write the letter.

squashing clay ●	→ A	● **A** changes its shape
catching a ball ●	→ C	● **B** starts it moving
pushing a swing ●	→ B	● **C** stops it
hitting a ball with a bat ●	→ D	● **D** changes its direction

3 Fill in the gaps. [3]

WORD BANK

gravity **friction** **pull** **light**

A mango falls down because of gravity. A ball stops on grass because of friction. A force is a push or a pull.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C2 · PUSHES AND PULLS

Forces all around

★★☆
Grade 2

Learning goals

- Explain friction and gravity.
- Compare forces.

1 Why is it easier to slide on a wet, smooth floor than on a rough mat? [2]

The smooth floor has less friction, so you slide; the rough mat has more friction.

2 Which needs a bigger force to push? Why? [2]



empty cart



cart full of heavy boxes

The full cart; heavier things need more force to move.

3 Draw an arrow to show which way gravity pulls a falling mango. [1]

Falling mango

An arrow pointing straight down.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · C2 · PUSHES AND PULLS

How far does the car roll?

★★☆
Grade 2

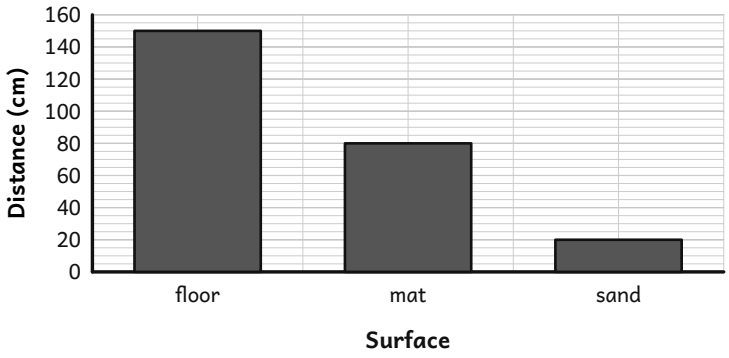
Learning goals

- Do a fair test.
- Measure distance.

Class 2B rolled a toy car down a ramp onto three surfaces and measured how far it went.

1 **Draw a bar chart.** [3]

Surface	smooth floor	mat	sand
Distance (cm)	150	80	20



2 **On which surface did the car go furthest?** [1]

the smooth floor

3 **Which surface had the most friction?** [1]

sand

4 **Why did they use the same ramp and car each time?** [1]

To make it a fair test.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C2 · PUSHES AND PULLS

The rickshaw van

★★★
Grade 2

Learning goals

- Find forces in real life.

Science | Work | Maths

Kamal drives a **rickshaw van** that carries vegetables to the market. He **pushes** the pedals with his feet to make it go. When the van is full of potatoes it is very **heavy**, and he must push much harder.

When he wants to stop, he squeezes the **brakes**. The brakes rub on the wheels and **friction** slows the van down. On a wet road the van slides more, so Kamal goes slowly.

A story about forces at work.

1 **What force does Kamal use to make the van go?** [1]

a push on the pedals

2 **Why is it harder to pedal when the van is full?** [2]

The van is heavier, so it needs a bigger force to move.

3 **How do the brakes stop the van?** [2]

They rub on the wheels; friction slows them down.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C2 · PUSHES AND PULLS

Show what you know

★★★
Grade 2

Learning goals

- Use force words.

1 Find the words.

O	T	E	O	U	O	G	E
U	O	M	T	L	Y	F	H
B	S	L	O	W	V	L	I
M	G	R	A	V	I	T	Y
E	F	D	F	P	U	L	L
C	A	I	F	O	R	C	E
I	S	S	P	U	S	H	C
E	T	B	T	E	O	H	O

fast force gravity pull push slow

2 Which force pulls things down? [1]

○ friction
● **gravity**
○ a push

3 Squashing clay changes its... [1]

○ colour
● **shape**
○ weight

4 Challenge. Draw yourself pushing something and pulling something. Label each. [2]

push
e.g. pushing a door

pull
e.g. pulling a rope